AI.209

Fuzzy Logic

COURSE OVERVIEW

Reasoning with imprecision: fuzzy sets, linguistic variables, and fuzzy control for uncertain systems.



ILLUSTRATIVE HERO IMAGERY

FUZZY LOGIC

1

SESSION

60 MINUTES

ORIENTATION: REASONING WITH VAGUENESS

KEY TOPICS

- Crisp vs. fuzzy sets
- Why classical logic is not enough
- Fuzzy logic in control
- Applications

TEACHING FOCUS

Model concepts that admit degrees.

KEY TAKEAWAY Many real-world properties are a matter of degree.

FUZZY LOGIC

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SESSION

60 MINUTES

FUZZY SETS & MEMBERSHIP

KEY TOPICS

- Membership functions
- Linguistic variables
- Fuzzy operators: AND, OR, NOT
- Fuzzification

TEACHING FOCUS

Quantify partial truth.

▲ **KEY TAKEAWAY** Fuzzy logic computes with words.

FUZZY LOGIC

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SESSION

60 MINUTES

FUZZY INFERENCE

KEY TOPICS

- Rule base: IF–THEN rules
- Mamdani and Sugeno inference
- Defuzzification
- Fuzzy rule design

TEACHING FOCUS

Derive crisp actions from fuzzy inputs.

KEY TAKEAWAY Rules plus inference make decisions.

FUZZY LOGIC

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SESSION

60 MINUTES

FUZZY CONTROL

KEY TOPICS

- Fuzzy PID
- Fuzzy control of nonlinear plants
- Tuning membership functions
- Stability considerations

TEACHING FOCUS

Control systems without precise models.

▲ **KEY TAKEAWAY** Fuzzy control works where models fail.

FUZZY LOGIC

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SESSION

60 MINUTES

FUZZY IN ROBOTICS

KEY TOPICS

- Fuzzy obstacle avoidance
- Fuzzy navigation
- Fuzzy sensor fusion
- Case studies

TEACHING FOCUS

Handle noisy, uncertain robot data.

▲ **KEY TAKEAWAY** Robots live in a fuzzy world.

FUZZY LOGIC

6

SESSION

 60 MINUTES

FUZZY LOGIC & REVIEW

KEY TOPICS

- Fuzzy controller implementation
- Course recap and assessment
- Toward deep learning (AI.210)
- Fuzzy vs. PID comparison

TEACHING FOCUS

Embrace uncertainty in control.

▲ **KEY TAKEAWAY** Precision is not always possible—or necessary.